THE STRUCTURE SHEAF OF A COUNTABLY INDEXED AFFINE FORMAL ALGEBRAIC SPACE HAS NO HIGHER COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof states that the general member of the site is treated in exactly the same way as the base object; the base case is carried out in full and the parallel case is not expanded, which is recorded as an obligation of the card.

Editorial note. The proof displays three families of exact sequences and the completed forms of the last two; all lie inside the counted span and are reproduced verbatim.

Editorial note. The statement carries a slogan environment in this chapter; the chapter's own preamble defines slogan as a comment environment, so such material is discarded at build time and is not part of the delivered mathematical text.

Editorial note. The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The Cech vanishing criterion on a site. Cohomology on Sites, Lemma sites-cohomology-lemma-cech-vanish-collection, invoked at proof lines 7771-7773. Reduces the vanishing of higher cohomology to the exactness of the Cech complex of a suitable family of coverings.
- The Cech complex for a covering by a single arrow. Etale Cohomology, Lemma etale-cohomology-lemma-cech-complex, invoked at proof lines 7775-7777. Gives the shape of the complex whose exactness has to be shown.
- The description of the affine etale site and of the values of the structure sheaf. Same chapter, Lemmas lemma-affine-identify-affine-etale and lemma-affine-etale-mcquillan, invoked at proof lines 7796-7799, together with the rule for the structure sheaf used at proof lines 7800-7804. Identifies the object of the site with a completion of an etale algebra and computes the structure sheaf on the iterated fibre products.
- A cover corresponds to a faithfully flat ring map, and the complex of a faithfully flat ring map is universally exact. Same chapter, Lemma lemma-etale-surjective at proof lines 7808-7810, and Descent, Lemma descent-lemma-ff-exact at proof lines 7810-7813. Supplies universal exactness of

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the uncompleted complex, which is what makes the topology comparison work.
- Completed short exact sequences remain exact in the countable case. Same chapter, Lemma lemma-ses, invoked at proof lines 7831-7833. Transfers exactness of the short exact sequences through completion, which is the conclusion.

Source reference. Original lemma 023M: sources/stacks-chow/tags.txt, source lines 2747–2747. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03OZ: sources/stacks-chow/tags.txt, source lines 4840–4840. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0DEF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7529–7529. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0DEE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7480–7480. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN9: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4492–4492. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AS0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 998–998. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *If X is a countably indexed affine formal algebraic space, then we have $H^n(X_{étale}, \mathcal{O}_X) = 0$ for $n > 0$.*

Proof. We may work with $X_{affine,étale}$ as this gives the same topos. We will apply Cohomology on Sites, Lemma 03F9 to show we have vanishing. Since $X_{affine,étale}$ has finite disjoint unions, this reduces us to the Čech complex of a covering given by a single arrow $\{U_{red} \to V_{red}\}$ in $X_{affine,étale} = X_{red,affine,étale}$ (see Étale Cohomology, Lemma 03OZ). Thus we have to show that

$$0 \to \mathcal{O}_X(V_{red}) \to \mathcal{O}_X(U_{red}) \to \mathcal{O}_X(U_{red} \times_{V_{red}} U_{red}) \to \ldots$$

is exact. We will do this below in the case $V_{red} = X_{red}$. The general case is proven in exactly the same way.

Recall that $X = \mathrm{Spf}(A)$ where A is a weakly admissible topological ring having a countable fundamental system of weak ideals of definition. We have seen in Lemmas 0DEE and 0DEF that the object U_{red} in $X_{affine,étale}$ corresponds to a morphism $U \to X$ of affine formal algebraic spaces which is representable by algebraic space and étale and $U = \mathrm{Spf}(B^\wedge)$ where B is an étale A-algebra. By our rule for the structure sheaf we see

$$\mathcal{O}_X(U_{red}) = B^\wedge$$

We recall that $B^\wedge = \lim B/JB$ where the limit is over weak ideals of definition $J \subset A$. Working through the definitions we obtain

$$\mathcal{O}_X(U_{red} \times_{X_{red}} U_{red}) = (B \otimes_A B)^\wedge$$

and so on. Since $U \to X$ is a covering the map $A \to B$ is faithfully flat, see Lemma 0AN9. Hence the complex

$$0 \to A \to B \to B \otimes_A B \to B \otimes_A B \otimes_A B \to \ldots$$

is **universally** exact, see Descent, Lemma 023M. Our goal is to show that

$$H^n(0 \to A^\wedge \to B^\wedge \to (B \otimes_A B)^\wedge \to (B \otimes_A B \otimes_A B)^\wedge \to \ldots)$$

is zero for $n > 0$. To see what is going on, let's split our exact complex (before completion) into short exact sequences

$$0 \to A \to B \to M_1 \to 0, \quad 0 \to M_i \to B^{\otimes_A i+1} \to M_{i+1} \to 0$$

By what we said above, these are universally exact short exact sequences. Hence $JM_i = M_i \cap J(B^{\otimes_A i+1})$ for every ideal J of A. In particular, the topology on M_i as a submodule of $B^{\otimes_A i+1}$ is the same as the topology on M_i as a quotient module of $B^{\otimes_A i}$. Therefore, since there exists a countable fundamental system of weak ideals of definition in A, the sequences

$$0 \to A^\wedge \to B^\wedge \to M_1^\wedge \to 0, \quad 0 \to M_i^\wedge \to (B^{\otimes_A i+1})^\wedge \to M_{i+1}^\wedge \to 0$$

remain exact by Lemma 0AS0. This proves the lemma. □